

SPECIAL TASK NO 2

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Domain: Artificial intelligence and machine Learning (AIML)

College Name: Manav School of Polytechnic AKOLA

Branch : Computer Engineering

GitHub Repositories Link : <https://github.com/anujaw193-star/python-project-collection>

- ❖ Load the dataset using Python (Pandas).

```
[1]
✓ 1s
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

plt.style.use("ggplot")
```

```
[2]
✓ 0s
df = pd.read_csv("anime.csv")
df.head()
```

	anime_id	title	score	rank	popularity	members	synopsis	start_date	end_date	type	episodes	image_u
0	28977	Gintama*	9.05	8	347	680037	Gintoki, Shinpachi, and Kagura return as the f...	2015-01-01	2016-01-01	TV	51.0	https://cdn.myanimelist.net/images/anime/3/720
1	57555	Chainsaw Man Movie: Reze-hen	9.18	2	962	282129	Sequel to Chainsaw Man .	2025-01-01	2025-01-01	Movie	1.0	https://cdn.myanimelist.net/images/anime/1763
2	11061	Hunter x Hunter (2011)	9.03	9	8	3107455	Hunters devote themselves to accomplishing haz...	2011-01-01	2014-01-01	TV	148.0	https://cdn.myanimelist.net/images/anime/1337
3	52991	Sousou no Frieren	9.29	1	121	1269074	During their decade-long quest to defeat the D...	2023-01-01	2024-01-01	TV	28.0	https://cdn.myanimelist.net/images/anime/1015
4	9969	Gintama'	9.02	11	406	601940	After a one-year hiatus, Shinpachi Shimura ret...	2011-01-01	2012-01-01	TV	51.0	https://cdn.myanimelist.net/images/anime/4/503

- ❖ Display the first few records.

```
[3]
✓ 0s
# Check Dataset Shape
print("Dataset Shape:", df.shape)
```

```
... Dataset Shape: (10000, 12)
```

❖ Display column names.

```
[4] # Display column names
print(df.columns)

... Index(['anime_id', 'title', 'score', 'rank', 'popularity', 'members',
'synopsis', 'start_date', 'end_date', 'type', 'episodes', 'image_url'],
dtype='object')
```

❖ Check data types

```
[5] # Check Data Types
df.dtypes

...
anime_id    int64
title       object
score       float64
rank        int64
popularity   int64
members     int64
synopsis     object
start_date  object
end_date    object
type        object
episodes    float64
image_url   object
dtype: object
```

❖ Identify missing values

```
[6] # Check Missing Values
df.isnull().sum()
```

	0
anime_id	0
title	0
score	0
rank	0
popularity	0
members	0
synopsis	2
start_date	2
end_date	84
type	0
episodes	48
image_url	0

dtype: int64

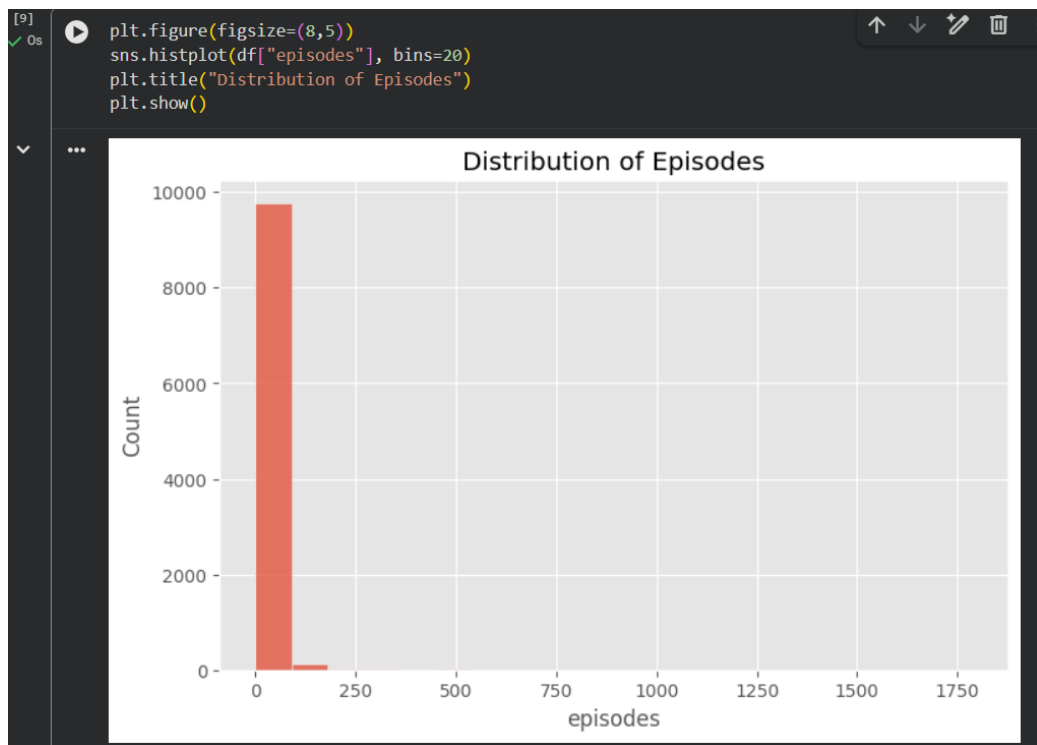
❖ Generate summary statistics.

```
[8] # Summary Statistics
df.describe(include='all')
```

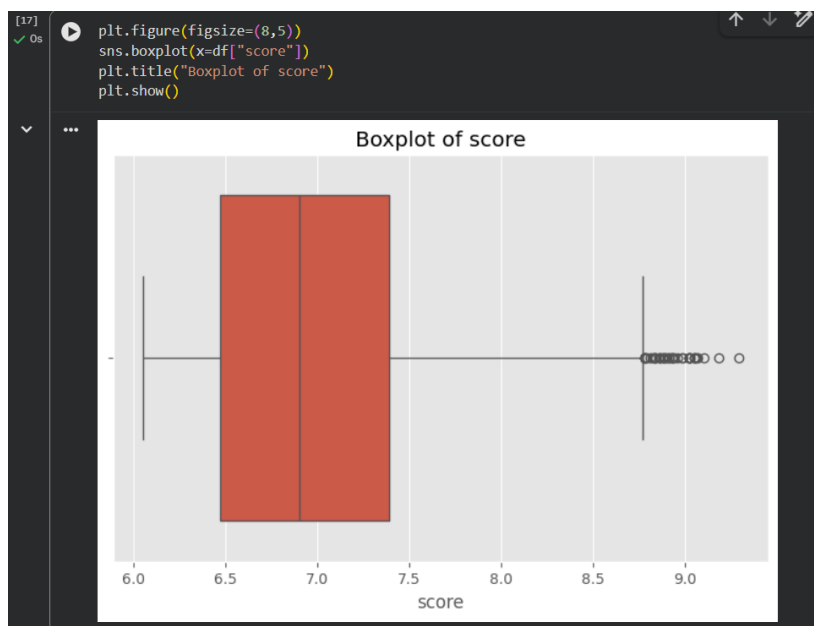
	anime_id	title	score	rank	popularity	members	synopsis	start_date	end_date	type	episodes
count	10000.000000	10000	10000.000000	10000.000000	10000.000000	1.000000e+04	9998	9998	9916	10000	9952.000000
unique	NaN	9999	NaN	NaN	NaN	NaN	9762	68	69	6	NaN
top	NaN	Ling Jian Zun 4th Season	NaN	NaN	NaN	NaN	No synopsis information has been added to this...	2023-01-01	2023-01-01	TV	NaN
freq	NaN	2	NaN	NaN	NaN	NaN	167	474	482	4330	NaN
mean	24129.128000	NaN	6.980589	5000.494000	6490.903700	1.075309e+05	NaN	NaN	NaN	NaN	15.702271
std	19948.188153	NaN	0.619406	2886.898727	4646.309341	2.733514e+05	NaN	NaN	NaN	NaN	41.865696
min	1.000000	NaN	6.050000	1.000000	1.000000	2.260000e+02	NaN	NaN	NaN	NaN	1.000000
25%	4055.500000	NaN	6.470000	2500.750000	2591.750000	4.270250e+03	NaN	NaN	NaN	NaN	1.000000
50%	21406.000000	NaN	6.900000	5000.500000	5564.500000	1.817900e+04	NaN	NaN	NaN	NaN	11.000000
75%	40028.750000	NaN	7.390000	7500.250000	9824.250000	8.469275e+04	NaN	NaN	NaN	NaN	13.000000
max	62863.000000	NaN	9.290000	10000.000000	22184.000000	4.262220e+06	NaN	NaN	NaN	NaN	1787.000000

VISUALIZATIONS:

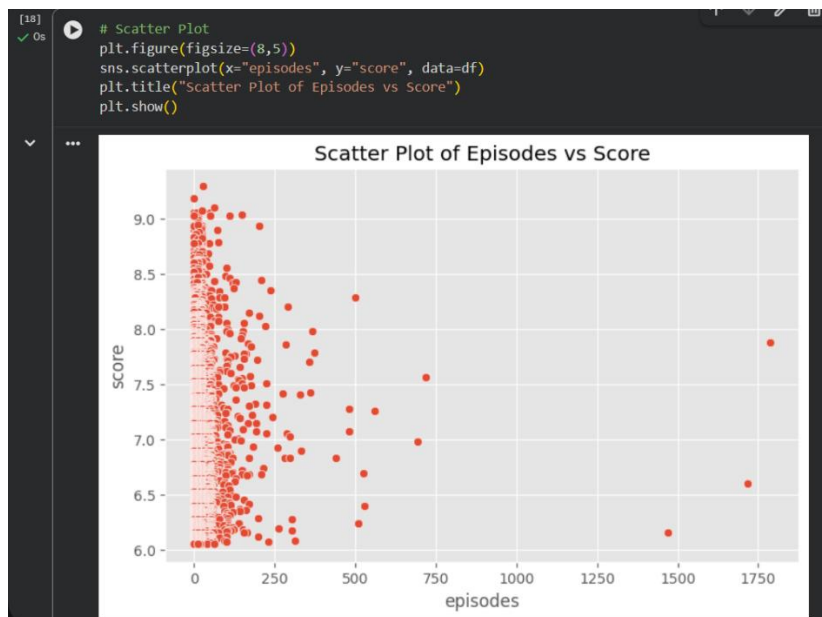
❖ Histogram :



❖ Box Plot

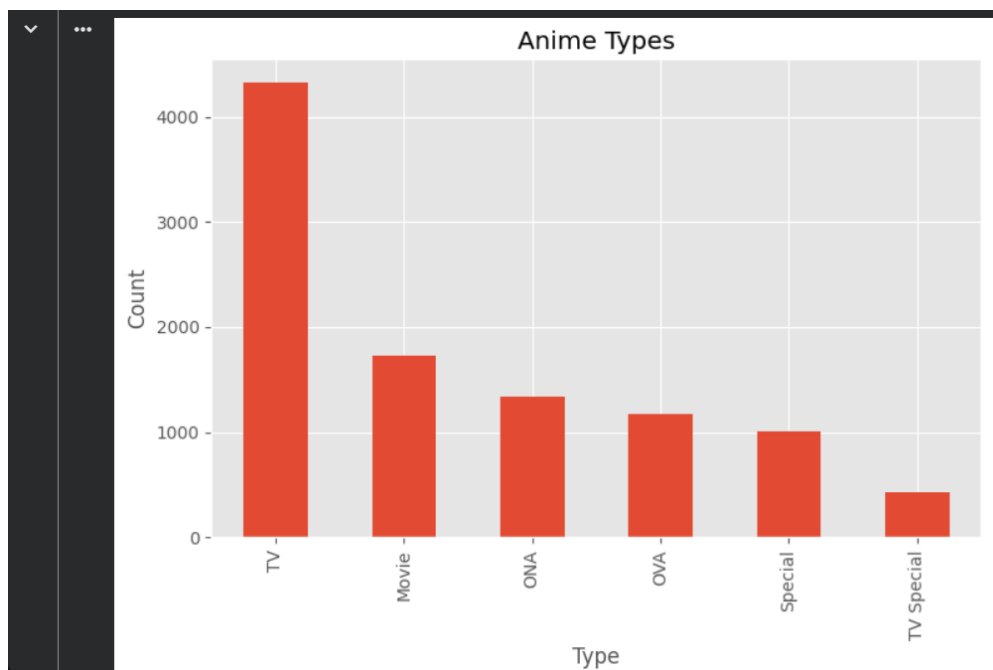


❖ Scatter Plot

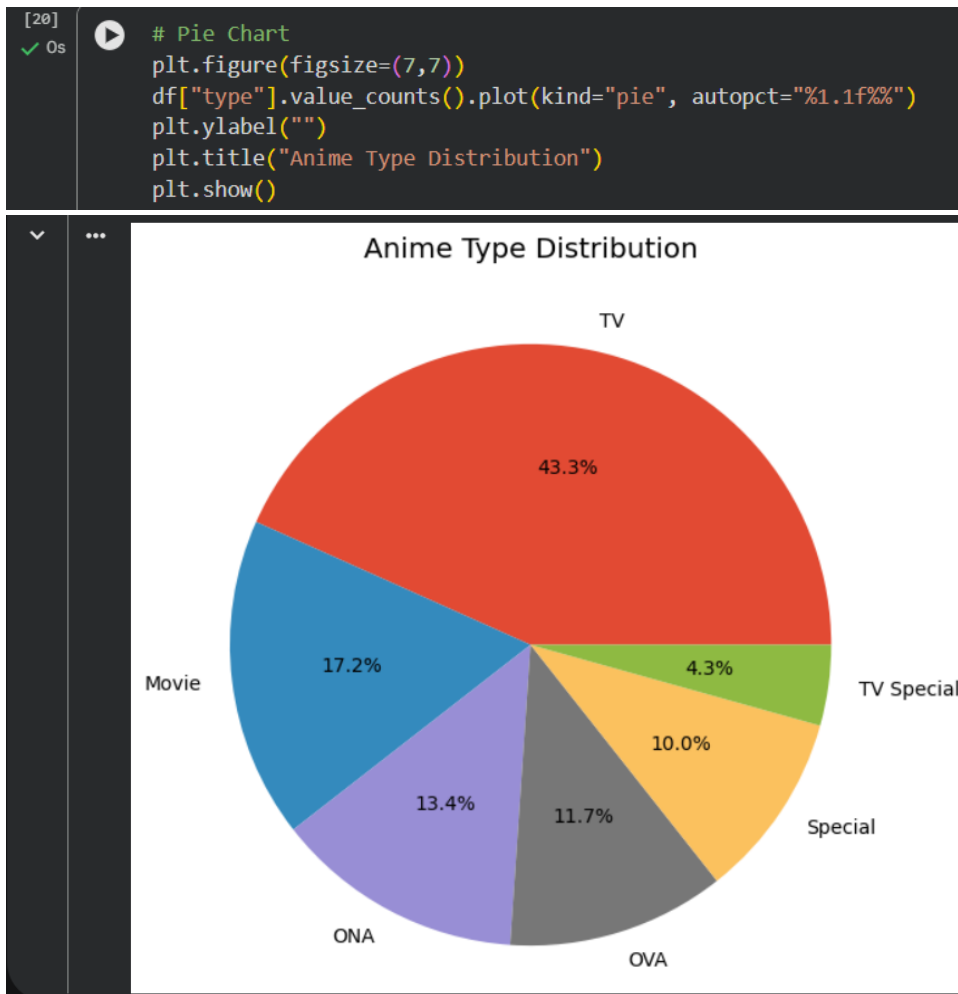


❖ Bar Chart

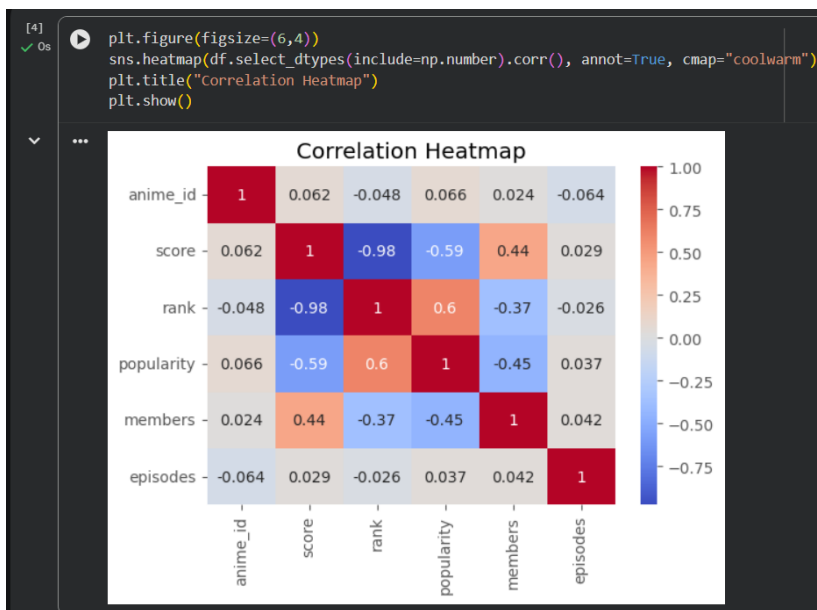
```
[19] # Bar Chart
plt.figure(figsize=(8,5))
df["type"].value_counts().plot(kind="bar")
plt.title("Anime Types")
plt.xlabel("Type")
plt.ylabel("Count")
plt.show()
```



❖ Pie Chart



❖ Heatmap



❖ Anime Dataset – Summary

The Anime dataset contains information about various anime titles, including their score, rank, popularity, number of members, type, number of episodes, release dates, and synopsis. The dataset was explored using Python libraries such as Pandas, Matplotlib, and Seaborn. Exploratory Data Analysis (EDA) was performed to understand the structure of the dataset, identify missing values, generate summary statistics, and visualize important patterns using different charts.

Key Insights

- The dataset contains information about different types of anime such as TV, Movie, OVA, ONA, and Special.
- The score column shows that most anime have ratings between 6 and 8.
- TV anime are the most common type compared to Movies, OVAs, and ONAs.
- Some anime have a very high number of episodes, which appear as outliers in the box plot.
- The heatmap shows the correlation between numerical features such as score, popularity, members, and episodes.
- A few columns contain missing values, which should be handled before building machine learning models.

❖ Dataset Name

Anime Dataset

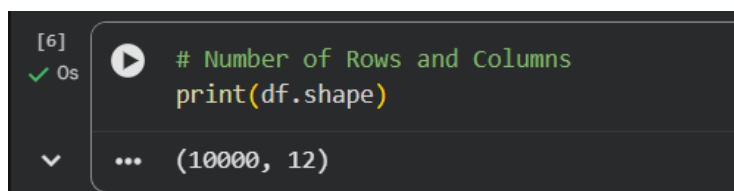
❖ Kaggle Dataset Link

<https://www.kaggle.com/datasets/udaykumar025/anime-dataset-top-10k-normalized>

❖ Dataset Description

The Anime dataset contains information about anime titles, including their names, scores, popularity, ranks, number of members, type, number of episodes, release dates, image URLs, and synopsis. It is commonly used for data analysis, recommendation systems, and machine learning projects.

❖ Number of Rows and Columns



```
[6] ✓ 0s # Number of Rows and Columns
print(df.shape)

... (10000, 12)
```

❖ List of Dataset Feature (Columns)

- anime_id

- title
 - score
 - rank
 - popularity
 - members
 - synopsis
 - start_date
 - end_date
 - type
 - episodes
 - image_url
-

❖ Input Features (Independent Variables)

- anime_id
- title
- rank
- popularity
- members
- synopsis
- start_date
- end_date
- type
- episodes
- image_url

❖ Output/ Target Variable (Dependent Variable)

Score

❖ Type of Machine Learning Problem

Regression

❖ Conclusion

The Anime dataset was successfully analyzed using Exploratory Data Analysis (EDA). The analysis helped identify the dataset structure, missing values, data types, and statistical information. Visualizations such as histograms, box plots, scatter plots, bar charts, pie charts, and heatmaps provided meaningful insights into anime scores, episode distribution, popularity, and anime types. The dataset is suitable for regression and recommendation system projects after appropriate data preprocessing and handling of missing values.

THE END